

Robot Framework — Task 4

1. Experiment Title

Assertions and Verification Using Robot Framework

2. Problem Statement

Write Robot Framework test cases that:

1. Perform an action and use an assertion to verify that the expected outcome matches the actual result.
2. Verify the response of an API request using the **Should Be Equal As Strings** keyword.

3. Objective

To demonstrate **assertions and verification techniques** in Robot Framework for both web application testing and API response validation.

4. Tools, Software and Concepts Used

Tools / Software

- Python
- Robot Framework
- SeleniumLibrary
- RequestsLibrary
- Google Chrome
- Visual Studio Code

Concepts

- Assertions
- Verification
- Selenium WebDriver
- API testing
- HTTP GET request
- Location Should Contain

- Title Should Be
- Should Be Equal As Strings
- HTTP status code verification
- Robot Framework variables

5. Source Code / Implementation Steps

File: assertions_test.robot

*** Settings ***

Library SeleniumLibrary

Library RequestsLibrary

*** Test Cases ***

Verify Webpage Result

Open Browser <https://www.saucedemo.com/> Chrome

Input Text id=user-name standard_user

Input Text id=password secret_sauce

Click Button id=login-button

Location Should Contain inventory.html

Title Should Be Swag Labs

Close Browser

Verify API Response

Create Session jsonplaceholder <https://jsonplaceholder.typicode.com>

`${response}= GET On Session jsonplaceholder /posts/1`

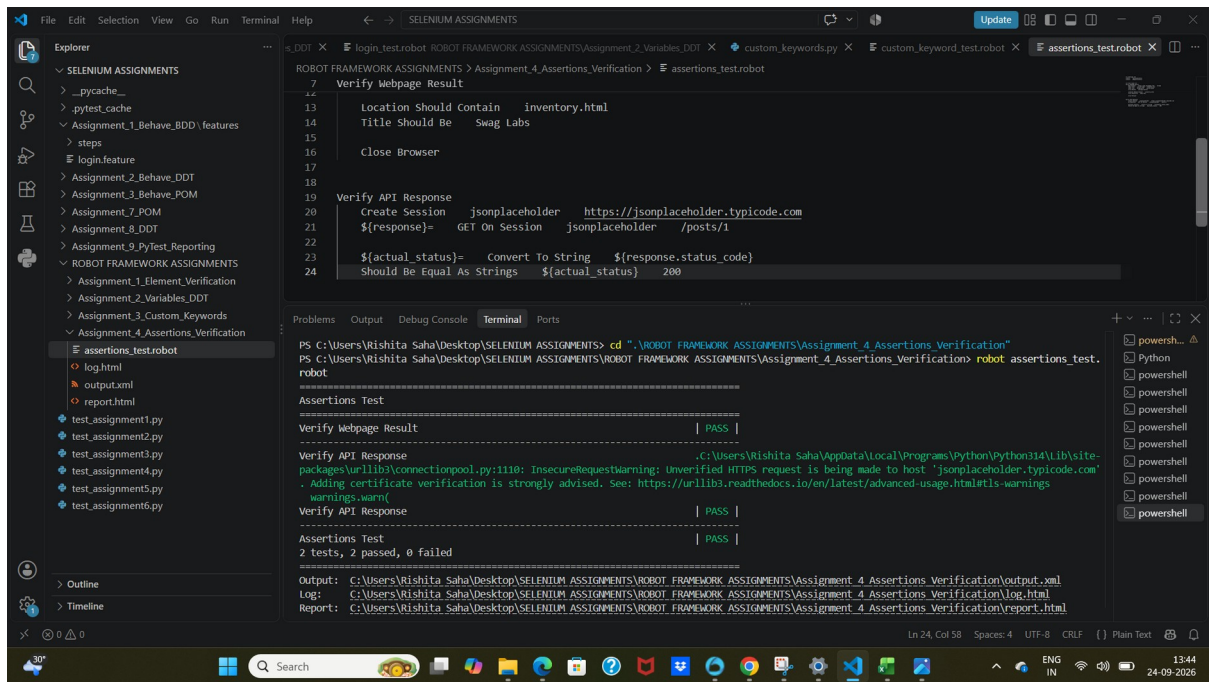
`${actual_status}= Convert To String ${response.status_code}`

`Should Be Equal As Strings ${actual_status} 200`

Implementation Steps

1. Import SeleniumLibrary for browser automation.
2. Import RequestsLibrary for API testing.
3. Open the SauceDemo login page.
4. Enter valid credentials and click the login button.
5. Verify that the URL contains inventory.html.
6. Verify that the page title is Swag Labs.
7. Create an API session with the JSONPlaceholder service.
8. Send a GET request to /posts/1.
9. Extract the HTTP status code.
10. Convert the status code to a string.
11. Use Should Be Equal As Strings to verify that the actual status is 200.

6. Output with Clearly Labelled Screenshots



Show the terminal with:

2 tests, 2 passed, 0 failed

Caption:

Figure 3: Successful execution of web and API assertion test cases

7. Result

The web application and API verification test cases were successfully executed. The SauceDemo login result and page title were verified, and the API response status code was successfully verified as 200.

8. Observation

The web assertions confirmed the expected URL and page title after login. The API test successfully retrieved the requested resource and verified its HTTP status code using Should Be Equal As Strings.

9. Conclusion

The experiment demonstrated how Robot Framework can be used to perform **assertions and verification for both web applications and APIs**. The test cases successfully validated expected results against actual outcomes.